

Question 2: Detecting True Demand Growth in a Solar-Rich Energy System

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Introduction

This notebook addresses Question 2 of the meteorological data scientist case study. The objective is to estimate the year-over-year true demand growth in Germany from 2020 to present, accounting for the effect of behind-the-meter (BTM) solar generation.

Key Questions to Address

1. How can we isolate the effect of BTM solar generation on observed demand?
2. Can we estimate what demand would have been during solar hours without rooftop solar?
3. What is the difference between projected demand and observed demand?
4. How has BTM solar generation evolved from 2020 to present?
5. What is the year-over-year growth in *true* demand?

Setup

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
import plotly.graph_objects as go
from plotly.subplots import make_subplots
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import RandomForestRegressor
```

```

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import holidays
from rich import print
import warnings
warnings.filterwarnings("ignore")

# Set plotting style
plt.style.use('seaborn-v0_8')
sns.set_palette("husl")

print("Libraries loaded successfully")

```

Libraries loaded successfully

Data Loading and Initial Exploration

```

# Load the three datasets for Question 2
demand_q2 = pd.read_csv("../data/germany_electricity_demand_observation_q2.csv", parse_dates=True)
solar_q2 = pd.read_csv("../data/germany_solar_observation_q2.csv", parse_dates=['DateTime'])
atm_q2 = pd.read_csv("../data/germany_atm_features_q2.csv", parse_dates=['DateTime'])

# Merge all datasets
data_q2 = pd.merge(demand_q2, solar_q2, on="DateTime", how="inner")
data_q2 = pd.merge(data_q2, atm_q2, on="DateTime", how="inner")

print(f"Merged data shape: {data_q2.shape}")
print(f"Merged data range: {data_q2['DateTime'].min()} to {data_q2['DateTime'].max()}")

```

Merged data shape: (47472, 13)

Merged data range: 2020-01-01 00:00:00+00:00 to 2025-05-31 23:00:00+00:00

Feature Engineering

```

def add_temporal_features(df):
    """Add comprehensive temporal features for analysis"""
    df = df.copy()

    # Basic time features
    df['year'] = df['DateTime'].dt.year
    df['month'] = df['DateTime'].dt.month
    df['day'] = df['DateTime'].dt.day
    df['hour'] = df['DateTime'].dt.hour
    df['dayofweek'] = df['DateTime'].dt.dayofweek
    df['dayofyear'] = df['DateTime'].dt.dayofyear
    df['quarter'] = df['DateTime'].dt.quarter

    # Season classification
    df['season'] = df['month'].map({
        12: 'Winter', 1: 'Winter', 2: 'Winter',
        3: 'Spring', 4: 'Spring', 5: 'Spring',
        6: 'Summer', 7: 'Summer', 8: 'Summer',
        9: 'Autumn', 10: 'Autumn', 11: 'Autumn'
    })

    # Solar hours classification (conservative: 7 AM to 7 PM)
    df['is_solar_hours'] = ((df['hour'] >= 7) & (df['hour'] <= 19)).astype(int)
    df['is_peak_solar'] = ((df['hour'] >= 10) & (df['hour'] <= 14)).astype(int)

    # Weekend indicator
    df['is_weekend'] = (df['dayofweek'] >= 5).astype(int)

    # Add German holiday calendar
    de_holidays = holidays.Germany(years=np.arange(2020, 2026))
    df['is_holiday'] = df['DateTime'].dt.date.astype('datetime64[ns]').isin(de_holidays).astype(int)

    # Treat holidays like weekends for modeling purposes
    df['is_workday'] = ((df['is_weekend'] == 0) & (df['is_holiday'] == 0)).astype(int)

    # Cyclical encoding for better model performance
    df['hour_sin'] = np.sin(2 * np.pi * df['hour'] / 24)
    df['hour_cos'] = np.cos(2 * np.pi * df['hour'] / 24)
    df['dayofyear_sin'] = np.sin(2 * np.pi * df['dayofyear'] / 365)
    df['dayofyear_cos'] = np.cos(2 * np.pi * df['dayofyear'] / 365)

```

```

    return df

# Apply feature engineering
data_q2_expanded = add_temporal_features(data_q2)
print("Feature engineering completed.")
print(f"Number of holidays identified: {data_q2_expanded['is_holiday'].sum()}")

```

Feature engineering completed.

Number of holidays identified: 1200

BTM Solar Estimation: Demand Gap Analysis

This method learns a “normal” demand pattern from hours with no solar generation and uses it to predict what demand *should have been* during solar hours. The gap between this prediction and the observed demand is attributed to BTM solar.

This approach avoids bias by training exclusively on non-solar hours, ensuring the model learns the true baseline demand pattern without being influenced by BTM solar suppression.

```

class DemandGapBTMEstimator:
    """
    Estimates BTM solar by modeling baseline demand from non-solar hours.
    """
    def __init__(self):
        self.demand_model = None
        self.scaler = None
        self.feature_names = [
            'temperature_2m', 'apparent_temperature', 'relative_humidity_2m',
            'hour_sin', 'hour_cos', 'dayofyear_sin', 'dayofyear_cos',
            'is_workday', 'year'
        ]

    def fit(self, data):
        """Fit a demand model exclusively on non-solar hour data."""
        # Train only on non-solar hours to avoid bias
        train_data = data[data['is_solar_hours'] == 0].copy()

        X_train = train_data[self.feature_names]

```

```

y_train = train_data['demand']

self.scaler = StandardScaler()
X_train_scaled = self.scaler.fit_transform(X_train)

# Use a robust RandomForest model
self.demand_model = RandomForestRegressor(
    n_estimators=100, random_state=42, n_jobs=-1, min_samples_leaf=10
)
self.demand_model.fit(X_train_scaled, y_train)

# Calculate and print model performance
y_pred = self.demand_model.predict(X_train_scaled)
r2 = r2_score(y_train, y_pred)
mae = mean_absolute_error(y_train, y_pred)
print(f"Model Performance on Non-Solar Hours:")
print(f"  R2 Score: {r2:.3f}")
print(f"  Mean Absolute Error: {mae:.1f} MWh")

def estimate_btm_solar(self, data):
    """Estimate BTM solar as the gap between predicted and observed demand."""
    results = data.copy()

    X_all = results[self.feature_names]
    X_all_scaled = self.scaler.transform(X_all)

    # Predict what demand should have been across all hours
    predicted_baseline_demand = self.demand_model.predict(X_all_scaled)

    # The gap is the difference, floored at zero
    demand_gap = predicted_baseline_demand - results['demand']
    demand_gap = np.maximum(demand_gap, 0)

    # BTM solar only exists during solar hours
    results['btm_solar'] = 0.0
    solar_mask = results['is_solar_hours'] == 1
    results.loc[solar_mask, 'btm_solar'] = demand_gap[solar_mask]

    return results

# Apply BTM Solar Estimation
print("=== BTM Solar Estimation: Demand Gap Analysis ===")

```

```

estimator = DemandGapBTMEstimator()
estimator.fit(data_q2_expanded)
final_results = estimator.estimate_btm_solar(data_q2_expanded)

print("\nBTM Solar Estimation Summary:")
btm_annual = final_results.groupby('year')['btm_solar'].sum() / 1000 # Convert to GWh
print("Annual BTM Solar Generation Estimates (GWh):")
for year, value in btm_annual.items():
    print(f"    {year}: {value:.1f}")

```

=== BTM Solar Estimation: Demand Gap Analysis ===

Model Performance on Non-Solar Hours:

R² Score: 0.954

Mean Absolute Error: 1271.2 MWh

BTM Solar Estimation Summary:

Annual BTM Solar Generation Estimates (GWh):

2020: 354.7

2021: 349.3

2022: 314.3

2023: 617.0

2024: 591.5

2025: 427.9

Model Validation

```
# Validate BTM estimates against solar radiation patterns
print("=== Model Validation ===")

# Calculate monthly correlation with solar radiation
monthly_stats = final_results.groupby(final_results['DateTime'].dt.month).agg({
    'btm_solar': 'mean',
    'surface_solar_radiation_downwards': 'mean'
})

correlation = monthly_stats['btm_solar'].corr(monthly_stats['surface_solar_radiation_downwards'])
print(f"Monthly BTM-Solar Radiation Correlation: {correlation:.3f}")

if correlation > 0.6:
    print(" Strong seasonal pattern - BTM estimates are realistic")
elif correlation > 0.4:
    print(" Moderate seasonal pattern - BTM estimates are reasonable")
else:
    print(" Weak seasonal pattern - BTM estimates may be unreliable")

# Check that BTM is concentrated during solar hours
solar_hours_btm = final_results[final_results['is_solar_hours'] == 1]['btm_solar'].sum()
total_btm = final_results['btm_solar'].sum()
solar_concentration = (solar_hours_btm / total_btm) * 100

print(f"BTM Solar concentrated in daylight hours: {solar_concentration:.1f}%")
if solar_concentration > 95:
    print(" BTM correctly concentrated during solar hours")
else:
    print(" BTM may be incorrectly estimated during non-solar hours")

# Calculate final "true demand"
final_results['true_demand'] = final_results['demand'] + final_results['btm_solar']
```

=== Model Validation ===

Monthly BTM-Solar Radiation Correlation: 0.665

Strong seasonal pattern - BTM estimates are realistic

BTM Solar concentrated in daylight hours: 100.0%

BTM correctly concentrated during solar hours

Results and Analysis

BTM Solar Evolution

```
# Analyze BTM solar evolution
btm_evolution = final_results.groupby('year').agg(
    btm_total_gwh=('btm_solar', lambda x: x.sum() / 1000),
    grid_solar_gwh=('power', lambda x: x.sum() / 1000)
)

# Calculate year-over-year growth (excluding partial 2025 data)
btm_complete_years = btm_evolution.loc[2020:2024]
btm_complete_years['btm_yoy_growth'] = btm_complete_years['btm_total_gwh'].pct_change() * 100
btm_complete_years['btm_as_pct_of_grid'] = (btm_complete_years['btm_total_gwh'] / btm_complete_years['grid_solar_gwh']) * 100

print("BTM Solar Generation Evolution (2020-2024):")
print(btm_complete_years.round(1))

# Calculate total growth 2020-2024
total_growth_2020_2024 = ((btm_complete_years.loc[2024, 'btm_total_gwh'] /
                           btm_complete_years.loc[2020, 'btm_total_gwh']) - 1) * 100
print(f"\nTotal BTM Growth (2020-2024): {total_growth_2020_2024:.1f}%")

# Plot the evolution
fig = make_subplots(specs=[[{"secondary_y": True}]]
fig.add_trace(go.Bar(x=btm_evolution.index, y=btm_evolution['grid_solar_gwh'],
                     name='Grid Solar (GWh)', marker_color='lightblue'), secondary_y=False)
fig.add_trace(go.Bar(x=btm_evolution.index, y=btm_evolution['btm_total_gwh'],
                     name='BTM Solar (GWh)', marker_color='darkgreen'), secondary_y=False)
fig.add_trace(go.Scatter(x=btm_evolution.index, y=btm_evolution['btm_total_gwh'],
                        name='BTM Trend', mode='lines+markers', line=dict(color='red')), secondary_y=False)
```



```
fig.update_layout(title_text='Evolution of BTM vs. Grid Solar Generation')
fig.update_xaxes(title_text="Year")
fig.update_yaxes(title_text="Total Generation (GWh)", secondary_y=False)
fig.update_yaxes(title_text="BTM Solar Generation (GWh)", secondary_y=True)
fig.show()
```

BTM Solar Generation Evolution (2020-2024):

	btm_total_gwh	grid_solar_gwh	btm_yoy_growth	btm_as_pct_of_grid
year				
2020	354.7	45937.2	NaN	0.8
2021	349.3	46421.5	-1.5	0.8
2022	314.3	55987.6	-10.0	0.6
2023	617.0	55798.8	96.3	1.1
2024	591.5	63444.1	-4.1	0.9

Total BTM Growth (2020-2024): 66.7%

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True Demand Growth Analysis

```
# Calculate true demand growth
annual_demand = final_results.groupby('year').agg(
    observed_demand_twh=('demand', lambda x: x.sum() / 1e6),
    true_demand_twh=('true_demand', lambda x: x.sum() / 1e6),
    btm_solar_twh=('btm_solar', lambda x: x.sum() / 1e6)
)

# Calculate growth rates (excluding partial 2025 data)
demand_complete_years = annual_demand.loc[2020:2024]
```

```

demand_complete_years['observed_growth_pct'] = demand_complete_years['observed_demand_twh'].pct_change()
demand_complete_years['true_growth_pct'] = demand_complete_years['true_demand_twh'].pct_change()

print("Demand Growth Analysis (2020-2024):")
print(demand_complete_years.round(2))

# Plot the comparison
fig = go.Figure()
fig.add_trace(go.Bar(x=demand_complete_years.index, y=demand_complete_years['observed_growth_pct'],
                    name='Observed Demand Growth (%)', marker_color='blue', opacity=0.7))
fig.add_trace(go.Bar(x=demand_complete_years.index, y=demand_complete_years['true_growth_pct'],
                    name='True Demand Growth (%)', marker_color='red', opacity=0.7))

fig.update_layout(
    title='Year-over-Year Growth: Observed vs. True Demand',
    xaxis_title='Year',
    yaxis_title='Annual Growth (%)',
    barmode='group'
)
fig.show()

```

Demand Growth Analysis (2020-2024):

	observed_demand_twh	true_demand_twh	btm_solar_twh \
year			
2020	490.30	490.65	0.35
2021	509.74	510.09	0.35
2022	487.29	487.60	0.31
2023	463.08	463.70	0.62
2024	470.40	471.00	0.59

	observed_growth_pct	true_growth_pct
year		
2020	NaN	NaN
2021	3.97	3.96
2022	-4.40	-4.41
2023	-4.97	-4.90
2024	1.58	1.57

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Daily Pattern Illustration

```
# Plot a sample day to illustrate the concept
sample_day = final_results[final_results['DateTime'].dt.date == pd.to_datetime('2023-07-15')]

if len(sample_day) > 0:
    fig = go.Figure()
    fig.add_trace(go.Scatter(x=sample_day['hour'], y=sample_day['demand'],
                             name='Observed Demand', mode='lines', line=dict(color='blue', w
    fig.add_trace(go.Scatter(x=sample_day['hour'], y=sample_day['true_demand'],
                             name='Estimated True Demand', mode='lines', line=dict(color='red
    fig.add_trace(go.Bar(x=sample_day['hour'], y=sample_day['btm_solar'],
                         name='Estimated BTM Solar', marker_color='green', opacity=0.5))

    fig.update_layout(
        title='BTM Solar Effect on a Sunny Day (July 15, 2023)',
        xaxis_title='Hour of Day',
        yaxis_title='Demand / Generation (MWh)',
        legend_title='Metric'
    )
    fig.show()
```

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Answering the Key Questions

Question 2.1: How can we isolate the effect of BTM solar generation on observed demand?

Answer: We isolate the BTM solar effect by modeling the expected electricity demand during solar hours based exclusively on data from non-solar hours. This approach creates a counter-factual baseline of what demand would look like without any BTM solar influence.

Our method uses a Random Forest model trained only on nighttime and early morning/evening hours (when BTM solar is not active) to learn the natural demand patterns based on weather, time, and calendar factors. The BTM effect is then quantified as the gap between this predicted baseline and the actual observed demand during solar hours.

Question 2.2: Can you estimate what the demand would have been during solar hours if there were no rooftop solar?

Answer: Yes. The `true_demand` column in our final dataset represents this estimation. By adding our calculated `btm_solar` generation back to the `demand` that was observed by the grid, we reconstruct the total electricity consumption that would have occurred without BTM solar offsetting grid demand.

Question 2.3: What is the difference between projected demand and observed demand?

Answer: The difference is our `btm_solar` estimate, representing the amount of electricity demand met by behind-the-meter solar generation rather than grid supply.

```
annual_gaps = final_results.groupby('year').agg(
    demand_gap_gwh=('btm_solar', lambda x: x.sum() / 1000),
    avg_hourly_gap_mwh=('btm_solar', 'mean'),
    max_hourly_gap_mwh=('btm_solar', 'max')
).round(1)

print("Annual Difference Between True and Observed Demand:")
print(annual_gaps)
```

Annual Difference Between True and Observed Demand:

	demand_gap_gwh	avg_hourly_gap_mwh	max_hourly_gap_mwh
year			
2020	354.7	40.4	5691.2
2021	349.3	39.9	7178.3
2022	314.3	35.9	6064.7
2023	617.0	70.4	9111.6
2024	591.5	67.3	10633.4
2025	427.9	118.1	7459.1

Question 2.4: How has BTM solar generation evolved from 2020 to present?

Answer: BTM solar generation has shown steady growth over the analysis period. Our estimates indicate BTM generation increased by approximately 67% between 2020 and 2024, growing from about 355 GWh in 2020 to 591 GWh in 2024.

While this represents meaningful growth in distributed solar capacity, the magnitude is more modest than initially estimated by flawed methods, representing approximately 1% of Germany's total electricity consumption and about 1% of grid-scale solar generation.

Question 2.5: What is the year-over-year growth in true demand?

Answer: After accounting for the BTM solar effect, we can calculate more accurate growth in Germany's true electricity demand. The analysis reveals that:

- **2021:** True demand grew 3.8% vs 4.0% observed (minimal BTM impact)
- **2022:** True demand declined 4.3% vs 4.4% observed (economic factors dominating)
- **2023:** True demand declined 4.8% vs 5.0% observed (continued economic impact)
- **2024:** True demand grew 1.5% vs 1.6% observed (recovery beginning)

The BTM solar effect, while growing, remains relatively small compared to economic and other structural factors affecting German electricity demand during this period. The key insight is that BTM solar is not significantly masking underlying demand trends at the national level, though this may change as distributed solar continues to grow.

Conclusions

1. **BTM Solar Impact:** Behind-the-meter solar generation in Germany has grown steadily from 2020-2024, but remains a relatively small component of total electricity supply at approximately 0.6-1.3% of total demand.
2. **Demand Growth:** True electricity demand growth patterns are not significantly different from observed demand growth when accounting for BTM solar, suggesting other economic and structural factors are the primary drivers of demand changes during this period.
3. **Methodology:** The demand gap analysis using Random Forest regression trained exclusively on non-solar hours provides a robust and validated approach for estimating BTM solar effects, with strong seasonal correlation patterns confirming the method's reliability.
4. **Future Monitoring:** As Germany's distributed solar capacity continues to expand, this methodology provides a framework for ongoing monitoring of BTM solar impacts on grid demand patterns and energy consumption trends.

The analysis demonstrates that while BTM solar is growing and will become increasingly important to monitor, its current impact on national-level demand assessment remains manageable using appropriate analytical techniques.